

Symmetry-resolved two-site entanglement as a key to Mott and pseudogap physics in the two-dimensional Hubbard model

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The SCES community approaches the third quantum revolution under a two-fold, bold assumption: strong inter-particle interactions grant **richer quantum resources** than what can be found in other platforms and these resources are **operationally accessible**. We are thus compelled to carefully assess these aspects, with appropriate measures of entanglement.

I will show how to identify the special entanglement properties of strongly-correlated electrons by symmetry-resolving the different quantum amplitudes connecting two sites in the lattice: one contribution is associated to antiferromagnetic fluctuations (*spin singlets*), another accounts for the spatial entanglement induced by the single-electron *hopping*, and the last term describes nonlocal *holon-doublon binding*.

The antiferromagnetic and holon-doublon contributions are directly connected to the charge and parity **superselection rules**. These fundamental constraints to fermionic entanglement have been repeatedly shown to faithfully identify the portion of correlations that can be harnessed in realistic quantum information protocols, other than reveal unique phenomena such as the spatial [1] and thermal [2] death of entanglement.

I will present **cluster dynamical mean-field theory** (CDMFT) results for salient landmarks in the phase diagram of the two-dimensional Hubbard model at zero temperature, relevant to the physics of unconventional superconductivity in copper oxides. The symmetry resolution of the two-site entanglement allows for a clear characterization of strong quantum correlations, both in the half-filled *Mott-Hubbard insulator* and in the *pseudogap metal* found at small hole-doping fractions. Antiferromagnetic *spin-singlet pairs dominate* the entanglement content of both the Mott-Hubbard insulator and the under-doped metal, at large U , while a significant *holon-doublon entanglement* is manifested *only in the intermediate coupling regime*, where the competition between U and t drives the Mott transition [3-5].

Remarkably, both contributions to the superselected two-site entanglement are suppressed by orders of magnitude in the *Fermi liquid regimes* found at weak interaction strengths and large doping fractions, whereas the *hopping entanglement remains large* [5].

These findings highlight the strong connection between superselection rules and the special quantum resources brought in by inter-particle interactions. Furthermore, they provide evidence of an RVB-like state for half-filled and underdoped cuprates, while also capturing explicitly the role of nonlocal holon-doublon binding in driving the Mott transition—an aspect traditionally considered beyond the reach of CDMFT approaches [6].

Finally, I will briefly showcase a recent **experimental verification** that spin-singlet (or charge superselected) entanglement is confined to nearest neighbors in the pseudogap phase of *cold-atom simulations* of the Hubbard model, and that it dies not only by overdoping the system, but also when crossing the pseudogap temperature T^* [7].

- [1] L. Ding, G. Dünneweber and C. Schilling., *Quantum Science and Technology* 9 (1), 015005, 2023
- [2] G. Bellomia*, F. Bippus*, G. Roósz, M. Capone and K. Held, *in preparation*
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- [4] G. Bellomia, *PhD Thesis, SISSA* (2024)
- [5] G. Bellomia, C. Mejuto-Zaera, M. Capone and A. Amaricci, *in preparation*
- [6] See, for instance, M. Capello et al., *Phys. Rev. Lett.* 94, 026406 (2005)
- [7] F. Bippus, T. Chalopin, G. Bellomia et al., *in preparation*